The market

Trunk section: §6.4 The market.

Talks to: Sources (§6.1: TradingView for index and sector-ETF quotes); Research data (§6.5: O'Neil's M and L, written nightly); the Card (§6.3), which carries the market context on every card; Registers (§6.6), whose R2 is a view of this domain's history; Learning (§6.7), which uses regime and sector as features.

Built on: MARKET-TAB (every block, formula and threshold of today's tab), CANSLIM §2 (the market model's exact rules), §8 (the tab's O'Neil block), §9 (the four levels of strength), §14 (the market reflected on each card), and the code in `src/sideD/` and `quant-platform/chart/oneil.py`, `groups.py`.

Contents

1. What the market domain answers
2. The legacy design in full
3. What was planned and never built
4. What failed
5. The new design
6. Contracts
7. How the rebuild is proven

---

1. What the market domain answers

Two different questions, kept apart on purpose:

Read	Answers	Clock	Built today by
The desk's read	what is the tape doing right now, and which sectors lead?	every scan	each tool, from TradingView quotes
O'Neil's method	should I be buying at all, and where is the leadership?	after each close (M also re-checked through the day)	qp, from index bars and whole-market RS

They use different words so the trader can never mistake one for the other. Both are stamped on every card, and both are frozen into the registers.

---

2. The legacy design in full

2.1 The inputs

TradingView scanner, every discovery scan, for **SPY, QQQ, IWM, DIA, VIX** and **15 sector ETFs**, with the columns `close`, `change`, `change_from_open`, `change|1W`, `SMA5`, `SMA20`, `SMA50`, `SMA200`, `BB.upper`, `BB.lower`, `ADX`, `VWAP`, `relative_volume_10d_calc`.

Sector	ETF		Sector	ETF
Technology	XLK		Real Estate	XLRE
Financial	XLF		Consumer Staples	XLP
Industrial	XLI		Semiconductors	SMH
Consumer Discretionary	XLY		Biotechnology	IBB
Energy	XLE		Retail	XRT
Health Care	XLV		Transportation	XTN
Communication Services	XLC		Utilities	XLU
Materials	XLB			

Yahoo daily bars of the **S&P 500 (^GSPC)** and the **Nasdaq Composite (^IXIC)** for O'Neil's M. **The SIC industry map** and **whole-market RS ratings** for L (both from the research worker: `ONEIL.md` (to write)).

No regime rather than a wrong one (D-S2): if SPY and QQQ are both missing any of close, SMA5, SMA20, SMA50 or SMA200, the scan records no regime, and cards carry no context.

2.2 Indexes

Five tiles in order SPY, QQQ, DIA, IWM, VIX: price, change today, change over the week. Tapping one opens its chart.

2.3 Short-term bias

From the index quotes (`computeMarketBiasDetail`):

Signal	+1	−1
SPY day	> +0.3 %	< −0.3 %
QQQ day	> +0.3 %	< −0.3 %
IWM day	> +0.3 %	< −0.3 %
VIX day	< −2 % (+1)	> +1 % (−1), > +3 % (−2)
SPY week	> +1 %	< −1 %
QQQ week	> +1 %	< −1 %

Sum ≥ **+3** BULLISH, ≤ **−3** BEARISH, else NEUTRAL. The tab shows the score and each signal.

2.4 Mid-term stage

From SPY (QQQ when SPY is incomplete) (`computeMarketStage`). Six checks, counted as bull signals:

1. close > SMA5
2. close > SMA20
3. SMA5 > SMA20
4. SMA20 > SMA50
5. “1-hour close > 20-hour MA”: the code uses the daily close > SMA20, which is check 2 again
6. “1-hour 5MA > 20-hour MA”: the code uses SMA5 > SMA20, which is check 3 again

Stage	Rule
UPTREND	≥ 5 bull, or 4 with close > SMA5
PULLBACK	3–4 bull, close < SMA5, SMA5 > SMA20
REBOUND	2–3 bull, close > SMA5, SMA5 < SMA20
SIDEWAYS	2–3 bull otherwise
DOWNTREND	fewer

Bollinger position (close between the bands): ≥ 75 % `UPPER` , ≤ 25 % `LOWER` , else `MID` .

2.5 Long-term bias

	SMA50 > SMA200	SMA50 < SMA200
close > SMA200	BULLISH	RECOVERING
close < SMA200	WEAKENING	BEARISH

Also shown: the distance from the 200-day in %.

2.6 The final regime

Long-term × mid-term, then a Bollinger overlay:

Long \ Mid	UPTREND	PULLBACK	REBOUND	SIDEWAYS	DOWNTREND
BULLISH	STRONG_UP	PULLBACK_BULL	UP	CHOP_BULL	CORRECTION
RECOVERING	WEAK_UP	RECOVERY	RECOVERY	BASING	DOWN
WEAKENING	RECOVERY	TOPPING	BEAR_RALLY	CHOP_BEAR	DOWN
BEARISH	BEAR_RALLY	DOWN	BEAR_RALLY	BASING	STRONG_DOWN

STRONG_UP at the upper band becomes **EXTENDED_UP**; STRONG_DOWN at the lower band becomes **CAPITULATION**.

Regime	Stance	Guidance shown
Extended Up	LONG	Market extended. Momentum strong. Pullbacks are buying opportunities.
Strong Uptrend	LONG	Strong bullish conditions. Favor long setups with momentum.
Uptrend	LONG	Bullish but recovering. Favor long setups selectively.
Weak Uptrend	LONG	Recovering from weakness. Light long bias.
Pullback (bull intact)	LONG	Short-term pullback in a bull market. Watch for long re-entries.
Recovery	NEUTRAL	Recovering structure. Be selective. Wait for confirmation.
Basing	NEUTRAL	Market is basing. No clear direction. Reduce exposure.
Chop (bull bias)	NEUTRAL	Choppy but leaning bullish. Trade small and be selective.
Chop (bear bias)	NEUTRAL	Choppy with bear undertone. Reduce position size.
Correction (bull intact)	NEUTRAL	Correction in a bull market. Monitor for stabilization before re-entering.
Topping	SHORT	Signs of topping. Reduce longs. Hedge or go short on weakness.
Bear Rally	NEUTRAL	Counter-trend rally in bear market. Fade strength cautiously.
Downtrend	SHORT	Bearish conditions. Avoid longs. Short on rallies.
Strong Downtrend	SHORT	Strongly bearish. Reduce all risk. Short only.
Capitulation	SHORT	Market in capitulation. Extreme caution. Watch for reversal signals.

Confidence is high when long and mid term agree in side, medium otherwise (shown as Aligned / Diverging).

2.7 Sector heatmap and score

Columns: sector, ETF, day %, week %, **RS day** (ETF day − SPY day), **RS week** (ETF week − SPY week), bias, score, ADX, status (🔥 HOT); sorted by score.

Score (`sectorShortTermBias`):

1. Five checks: close > SMA20 · SMA20 > SMA50 · close > VWAP · RS day > 0 · week > 0.
2. `base = checks / 5 × 100`.
3. `RS influence = RS day × 5`, clamped to ±30.
4. ADX factor: **1.3** if ADX > 25, **1.1** if 20–25, **0.8** below 20.
5. `score = round((base + RS influence) × ADX factor)`, clamped to ±100.
6. Bias: > **20** BULLISH, < **−20** BEARISH (`sectorBullishThreshold` , `sectorBearishThreshold`), else NEUTRAL.

2.8 Hot sectors

Counted **per New York trading day**, not per scan (F-17):

- **hot at once** when the day's score reaches **60** (`hotImmediateThreshold`);
- **hot after 3 trading days** (`hotSustainedSessions`) whose best score reached **40** (`hotSustainedThreshold`);
- **stays hot** while the day's best score is at least **20** (`hotFloorThreshold`); after **2** days below (`coolOffDays`) it cools off.

The state lives in each tool's memory; a restart starts it again; Settings has a reset.

2.9 Each card's sector

A card's TradingView sector is mapped to one of the 15 (`broadResolved`):

TradingView sector (what the stock scanner sends)	Market sector
Electronic Technology, Technology Services	Technology
Finance	Financial
Finance/Real Estate	Real Estate
Producer Manufacturing, Process Industries	Industrial
Consumer Services, Consumer Durables	Consumer Discretionary
Consumer Non-Durables	Consumer Staples
Energy Minerals	Energy
Health Technology, Health Services	Health Care
Communications	Communication Services
Non-Energy Minerals	Materials
Retail Trade	Retail
Transportation, Utilities	themselves

GICS-style names (Healthcare, Consumer Cyclical, Basic Materials, Consumer Defensive, …) are mapped too, for the index scanner. Then each card carries `secBias`, `secScore` and `secHot` (🔥 HOT) of its sector.

2.10 M: O’Neil’s market model

Rebuilt from O’Neil’s published descriptions (CANSLIM §2):

Rule	Value
Distribution day	close down **≥ 0.2 %** on volume **above the prior session**, counted only in an uptrend
Stalling day	heavy volume (above the prior session or the 25-day average), close in the **lower half** of the range, gain **under 0.2 %**: counts as distribution
Live window	**25 sessions**
Removal	after 25 sessions, or when the index trades **5 %** above that day's close
Under pressure	**5** live distribution days
Correction	**6** live, or the uptrend's low broken
Rally attempt, day 1	the session after the **lowest close** of the correction; a lower close restarts the count
Follow-through day	day **4 or later**, gain **≥ 1.7 %**, volume above the prior session; "late" after day 7, "very late" after day 11 (still counted)

Volume above the 50-day average is reported beside a follow-through but does not decide it. Each index runs its own state machine; **the published status is the worse of the two**:

Status	What O'Neil does (shown)
Confirmed uptrend	Buy breakouts. This is the state the method is designed for.
Uptrend under pressure	Stop initiating new positions; tighten stops. A name holding up through distribution is a leader candidate for the next follow-through.
Market in correction (including a rally attempt with no follow-through yet)	No new buying. Three out of four stocks follow the market.

The block shows: the status, its date and a *because* line; live distribution per index; the follow-through (date, day number, gain, timing); sessions since it, with a band; the rally attempt (day N off a dated low); the guidance; a **dated distribution-day table** (date, index, kind, move %, volume ratio, sessions until expiry); the rules in use; the caveat that about one follow-through in four fails; the model note.

On each card (CANSLIM M and Market Context): the status, live distribution days, what this stock did on those days (`up on 4 of 5, avg +0.9 % vs index −0.8 %`), the verdict HOLDING UP / IN LINE / GIVING WAY, sessions since the follow-through, and the exposure note.

2.11 L: industry groups

Ranked across the whole market, the level O'Neil says carries about **37 %** of a stock's move (the sector **12 %**):

Step	How
Membership	SIC codes from EDGAR for every filer; TradingView labels only for symbols SIC has not classified (F-21)
Strength	each stock’s RS rating, ranked against the whole market
Group strength	the **median** member RS
Thin groups	fewer than **6** ranked members: merged one rung up (4-digit industry → 3-digit industry group → 2-digit major group); dropped if still thin
Ranking	1 … N by median RS; percentile and letter (A+ ≥ 94 … E)
Rotation	rank three months ago → now: into / out of leadership

The table: top 20 and weakest 10, each with rank X of N, group (marked *rolled up*), percentile and letter, members, median RS, share over RS 80, the 3-month rank change, and the four leaders with their RS. A note gives how many symbols and industries are mapped and how many groups are ranked.

2.12 Who uses what

Number	Card	R1	R2	Model features
Regime, long / mid / short	Market Context	✓	✓	✓
Sector bias, score, hot	Market Context, 🔥 HOT	✓	✓ (all 15)	bias and score
Index prices	—	—	✓	—
O’Neil status	CANSLIM M	`csMarket`	✓ (read at export)	—
Group rank	CANSLIM L	`csGroup*`	—	—

---

3. What was planned and never built

From CANSLIM §9 and §12.1:

Item	What it is	In the new design
Index RS line (M11)	`Nasdaq ÷ S&P 500`, the earliest read on whether growth or defensives are being bought	built: a line on the tab
Sector rank 1 of 33 (L7)	a coarse level between the 15 ETFs and the groups	built from the SIC map’s major groups
Group new highs / lows (L4)	how many members made new highs or lows	built
Divisor stored with every group rank (L2)	a group scheme can change (IBD went from 197 to 145 groups); a stored rank without its divisor rewrites its own past	built: every rank is stored as `rank of N`

4. What failed

ID	Fault	Root cause	Prevented by
—	five tools compute the same snapshot from the same ETFs, minutes apart, and can disagree	the market read lives inside each tool	one market engine
F-17 (fixed) / —	hot state was once counted per scan; it is still lost on every restart	state in memory	hot state persisted per trading day
F-16	sector score runs about −39 to +100, so BULLISH is far easier to reach than BEARISH	`base` runs 0–100 before the RS influence	score centred on 0 (§5.4)
D-D5	the mid-term stage counts two facts twice	“hourly” checks proxied by daily ones	four distinct checks, or real hourly data
D-D3	R2’s O’Neil columns are read when exported, so an old day shows today’s status	O’Neil status not frozen with the snapshot	R2 reads the market history, which holds the O’Neil status as it was
D-D6	the group table marks only a two-rung roll-up	display	one rule for both rungs, as on the card
D-S2 (fixed)	no index data once read as a bearish market	no honest null	kept: no regime, said plainly
new	four TradingView stock sectors have no mapping: Commercial Services, Distribution Services, Industrial Services, Miscellaneous. Their cards get no sector bias, score or hot badge	an incomplete map	a complete, tested map; unmapped labels reported in health
—	when an O’Neil file is missing, the tab asks qp to build it inside a page request	computation in the request path	the tab only reads; a missing fact says “not built yet · last run DATE”

5. The new design

5.1 One market engine

The market engine runs in desk-worker. At every discovery scan (and at least every 5 minutes during 04:00–16:00), it:

1. asks the TradingView adapter for the 5 indexes and 15 sector ETFs, once;
2. computes the desk’s read (§5.4);
3. reads the current O’Neil facts (M, L) from the fact store;
4. writes one **market state** row with everything above, stamped with its instant;
5. updates the persisted **hot-sector state** for the trading day.

Every card is stamped with the **id** of the market state it used, so the context on a card can always be traced to one snapshot.

5.2 The market state

Field group	Contents
`at`	the instant (UTC); the trading day (New York)
`indexes`	for SPY, QQQ, IWM, DIA, VIX: every column of §2.1
`sectors`	for each of the 15 ETFs: its columns, RS day, RS week, checks, score, bias, ADX factor, hot
`short_term`	the six signals, the sum, the bias
`mid_term`	the checks, the stage, Bollinger position
`long_term`	the bias, distance from the 200-day
`regime`	slug, label, stance, guidance, confidence
`oneil_m`	status, why, per-index live counts, follow-through, rally attempt, the distribution days, the rules in use: copied from the latest M fact so the state is complete on its own
`oneil_l_ref`	the id of the group ranking current at that instant
`status`	`ok`, or `no_regime` with the reason

History is kept: R2 and the backtest read it, and a day can be replayed.

5.3 The hot-sector state

One row per sector per trading day: the day’s best score, consecutive days at 40 or more, consecutive days under 20, hot or not, and since when. Settings keep the thresholds (60, 40, 3, 20, 2). A restart reads the rows back; a reset is an explicit action that is logged.

5.4 The desk read, rebuilt

The formulas of §2.3–§2.8 are kept, with three corrections that need a re-tune, each behind a versioned definition so history records which ran:

Correction	Change	Re-tune
Sector score centred on 0 (F-16)	`base = (checks − 2.5) / 2.5 × 100` (−100 … +100) before the RS influence and ADX factor	the hot thresholds (60, 40, 20) and bias thresholds (±20), checked against a month of history
Mid-term stage (D-D5)	four distinct checks (close > SMA5, close > SMA20, SMA5 > SMA20, SMA20 > SMA50), or hourly bars when an adapter provides them	the stage rules, re-cut for four checks
The sector map	every TradingView sector mapped; an unknown label is reported, never silently dropped	—

5.5 O’Neil on the tab

M and L are **computed by the research worker** (`ONEIL.md` (to write)) and **only read** here. The rules in §2.10 and §2.11 carry over unchanged. Through the day, the market engine re-checks M against the latest index quote (a distribution day in progress), labelled as provisional until the close.

5.6 The tab

Order, top to bottom:

1. **M · O’Neil market model** (first, because it decides whether to buy at all), with the index RS line.
2. **Indexes.**
3. **Desk analysis**: short-term, mid-term, long-term, the final regime, titled “Desk regime” so it can never be read as O’Neil’s status.
4. **Sector heatmap**, with the sector rank above it.
5. **L · Industry groups**: top 20, weakest 10, with divisors, roll-up marks, new highs and lows.

Every number has its ⓘ (what it means, how it is calculated, what it is not, source and as-of), as on the card.

5.7 What the card receives

`market_state_id`, `regime`, `regimeLabel`, `longTerm`, `midTerm`, `shortTerm`, `broadResolved`, `secBias`, `secScore`, `secHot`, and from M the per-card reflection (status, live distribution days, this stock on those days, verdict, sessions since the follow-through). See CARD.md §2.9 and §2.11.

6. Contracts

Owns

Table	Key	Written
`market_states`	id; instant	each scan
`hot_sector_days`	sector, trading day	each scan

Reads

From	What
sources	TradingView index and ETF quotes
research data	the latest `oneil_m` fact and group ranking

Read by

Domain	What
card	the current market state
registers	R2 at the tool’s slots
learning	regime and sector features
quant	the market state as of any day (backtest filters, the CANSLIM gate)

Routes (desk-web)

Route	Returns
`GET /api/market/state?at=`	the market state at an instant (default: now)
`GET /api/market/history?day=`	every state of a trading day
`GET /api/market/groups?day=`	the group ranking
`POST /api/market/hot/reset`	reset hot state (logged)

7. How the rebuild is proven

1. **Parity.** For a week the market engine runs beside the old tools; each scan, its regime, stages and sector scores are compared with the old tool’s snapshot (before the corrections of §5.4 are switched on).
2. **Fixtures.** Recorded TradingView answers drive every formula in CI, including incomplete SPY and QQQ (no regime).
3. **Restart test.** Restart desk-worker: hot state and the day’s history are intact.
4. **M against charts.** The dated distribution-day table is checked by hand against index charts for one month, as the legacy design requires.
5. **Sector map coverage.** A health check reports any card sector with no mapping.